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Abstract

Multislice simulation is the de-facto method for simulating the electron beam interaction with a sample in transmission electron microscopy (TEM). The workhorse of the multislice method is the Fresnel propagator, which is justified as a propagation solution of the paraxial Schrodinger equation. However, in cases such as when the sample or detector is tilted to a large angle relative to the electron beam, the paraxial approximation may no longer be sufficiently accurate. We explore alternative solutions based around the non-paraxial Schrodinger equation solved via the Rayleigh Sommerfeld propagator for single slice and arbitrary sample and detector tilts, the angular spectrum method for arbitrary sample tilts, and the Wave Propagation Method.

1 Introduction

For the simulation of typical beam-sample interactions (perpendicular plane wave TEM illumination, or STEM probe illumination) in the electron microscope, the traditional multislice propagation method has become the go to workhorse [?, ?]. It's range of accuracy of approximately 7 degrees stems from the fact that the Fresnel Propagator which solves free space propagation in the multislice method is itself derived from a Taylor expansion of the non-paraxial free space propagation kernel of the Rayleigh Sommerfeld diffraction integral. The Taylor expansion of the Rayleigh Sommerfeld propagator converts spherical waves to parabolic waves, whose accuracy degrades with increasing angle from the optical axis, and at 7 degrees the error is approximately X% (*citation needed*), deemed outside the range of acceptable accuracy for most electron microscopy applications. Therefore, most forward multislice simulations assume small angle scattering.

In section one, we explore alternative propagation solutions that are accurate beyond the paraxial approximation. The first non-paraxial method relies on a Fourier space solution to the Rayleigh Sommerfeld diffraction integral, the angular spectrum of plane waves. The second method relies upon the wave propagation method, which is an operator solution to the Helmholtz wave equation, and is accurate up to 85 degrees. Its drawbacks are that not for each unique refractive index (potential) value within the sample, a unique propagator, and thus unique Fourier transform must be performed, making it computationally expensive for samples with a wide range of refractive index or potential values.

2 Relativistic Dirac Equation and the approximated Klein-Gordon Equation

As described by [?], the relativistic Dirac equation is the starting point for the most accurate description of high-energy electron scattering:

$$E\Psi(\vec{r}) = [c\vec{\alpha} \cdot \hat{p} + mc^2\beta + V(\vec{r})]\Psi(\vec{r}), \quad (1)$$

where E denotes the energy, c the velocity of light, m the rest mass of the electron, V the electrostatic potential, $\vec{\alpha}$ and β the Dirac matrices, $\hat{p}$ the momentum operator and $\vec{r}$ the three-dimensional space coordinate. A solution to equation (1) was used by [?] as a "ground-truth" numerical baseline to establish the accuracy of the multislice solution to the relativistically corrected Schrodinger equation. In their work, they demonstrated that:

1. The spin effects of the dirac equation are mostly negligible to the error in amplitude of a central and high angle beam scattered through a thick sample of gold.

2. The taylor expansion that is applied to the $E^2 - V^2$ term, to enable one to use the simple relationship $\sigma * V$, where σ is the so called interaction constant given by:

$$\sigma = \frac{2\pi m e \lambda}{h^2}$$

does not contribute a significant error to the solution of the wave equation.

However, for a diffracted beam at 135 mrad, and scattering through a thick sample of gold, they found a divergence between the multislice solution solved via a split step fourier method with the Fresnel propagator, and a "ground-truth" ODE solution to the approximated Klein-Gordon (a.k.a relativistically corrected Schrodinger Wave Equation), Klein-Gordon, and Dirac equation. They explain the discrepancy by the fact that the multislice solution linearises the second order term in the wave equation, $d^2\psi/dz^2$, by assuming that the wave function can be written as a slowly varying envelope modulating a fast oscillating term along the beam propagation direction z . Furthermore, we note that the multislice solution ignores the change of wavelength of the electron as it propagates through a potential $V(\mathbf{r})$, since MS splits up the potential interaction and free space propagation steps. We cannot also overcome these limitations of the multislice method by just solving the wave equation in an ODE form, as it normally requires a periodic potential, and the slow speed of an ODE solution to the wave equation does not lend itself to repeated simulations over many frozen phonon configurations to include thermal vibrations, which has become a standard requirement to include thermal diffuse scattering in TEM simulations. A method that is both accurate to high angles, and can leverage the efficiency of the fast fourier transform is thus desirable to overcome these limitations.

In order to investigate non-paraxial solutions to the relativistically corrected Schrodinger equation that can leverage the efficiency of the fast fourier transform, it is insightful to write the Dirac, Klein-Gordon, and approximated Klein-Gordon in the same framework as a Time-Independent Helmholtz like wave equation. which is of the form:

$$\nabla^2 u + k^2 n^2(\mathbf{r}) u = 0. \quad (2)$$

with u the wave function, k the wave number in vacuum, and $n(\mathbf{r})$ the refractive index distribution of the medium.

Starting from the Time Independent Dirac equation:

$$\left[c\vec{\alpha} \cdot \hat{\vec{p}} + \beta mc^2 + V \right] \psi = E\psi \quad (3)$$

by applying the operator $(c\vec{\alpha} \cdot \hat{\vec{p}})$, and performing some algebraic manipulations, we arrive at:

$$\nabla^2 \psi + k_0^2 n^2(\mathbf{r}) \psi = -\frac{i}{\hbar c} (\vec{\alpha} \cdot \nabla V) \psi \quad (4)$$

where

$k_0 = \frac{\sqrt{E^2 - m^2 c^4}}{\hbar c}$ represents the vacuum wave number, and $n(\mathbf{r}) = \sqrt{\frac{(E-V)^2 - m^2 c^4}{E^2 - m^2 c^4}}$ is the effective refractive index of the medium.

From this Helmholtz like representation, it is straight forward to arrive at the Klein-Gordon equation by setting the α matrices to zero ignoring the effects of spin,

$$\nabla^2 \psi + k_0^2 n^2(\mathbf{r}) \psi = \nabla^2 \psi + \frac{(E - V)^2 - m^2 c^4}{\hbar^2 c^2} \psi = 0. \quad (5)$$

The relativistically correct Schrodinger equation arises from the Klein-Gordon equation by assuming that the potential V is small compared to the total energy E of the electron, and that when $(E - V)^2$ is expanded, the V^2 term can be neglected giving:

$$\nabla^2 \Psi + \left[\frac{E^2 - m^2 c^4}{\hbar^2 c^2} - \frac{2\gamma m V}{\hbar^2} \right] \Psi = 0 \quad (6)$$

Finally, we can write [?] with the usual interaction constant $\sigma = \frac{2\gamma m}{\hbar^2 k_0}$ as:

$$\nabla^2 \Psi + k_0^2(1 - \sigma V)\Psi = 0 \quad (7)$$

Writing equation (1) in its Helmholtz form (4) allows us clearly to see the relationship between the Time Independent Dirac equation and its approximations, and the classical Helmholtz wave equation in optics. Understanding the connections in this way allows us to directly apply non-paraxial propagation methods from optics community to electron microscopy.

Equation	Helmholtz Form	Refractive Index ($n(\mathbf{r})$)
Dirac	$\nabla^2 \psi + k_0^2 n^2 \psi = -\frac{i}{\hbar c}(\vec{\alpha} \cdot \nabla V)\psi$	$\sqrt{\frac{(E-V)^2 - m^2 c^4}{E^2 - m^2 c^4}}$
Klein-Gordon	$\nabla^2 \psi + k_0^2 n^2 \psi = 0$	$\sqrt{\frac{(E-V)^2 - m^2 c^4}{E^2 - m^2 c^4}}$
Rel. Schrödinger	$\nabla^2 \psi + k_0^2 n^2 \psi = 0$	$\sqrt{1 - \sigma V}$

Table 1: Helmholtz versions of quantum mechanics equations. The vacuum wave number is defined as $k_0 = \frac{\sqrt{E^2 - m^2 c^4}}{\hbar c}$.

3 Wave Propagation Method

The Wave Propagation Method (WPM) is a numerical method for solving the Helmholtz wave equation in the forward scattering approximation, that was introduced by [?], and has been demonstrated to be accurate up to 85 degrees from the optical axis [?]. A systematic derivation of the wave propagation method from the Helmholtz wave equation is given by [?]. For conciseness we present a summary of the derivation here:

Starting with the scalar Helmholtz equation for the wavefunction ψ in a medium with a z-dependent refractive index n :

$$(\partial_z^2 + \nabla_\perp^2 + k_0^2 n^2(x, y, z))\psi = 0$$

By neglecting back-scattering (the "one-way" wave approximation), the evolution of the wavefunction is governed by the square-root operator:

$$\partial_z \psi = i\sqrt{\nabla_\perp^2 + k_0^2 n^2} \cdot \psi$$

which has a solution according to:

$$\psi(x, y, z + \Delta z) = e^{i\sqrt{\nabla_\perp^2 + k_0^2 n^2} \Delta z} \psi(x, y, z)$$

From this point, one can derive the typical multislice solution (denoted Beam Propagation Methods (BPMs) in optics literature) by applying various approximations to the square root operator. Standard multislice approximates this square root by splitting the Laplacian ($\nabla_\perp^2$) and refractive index (n^2) terms. WPM, however, solves the propagation step Δz using a more rigorous Fourier Integral formulation:

$$\psi(x, y, z + \Delta z) = \frac{1}{2\pi} \iint \tilde{\psi}(k_x, k_y, z) e^{i\sqrt{k_0^2 n^2(x, y) - k_\perp^2} \Delta z} e^{i(k_x x + k_y y)} dk_x dk_y \quad (8)$$

Here, $\tilde{\psi}$ is the Fourier transform of the wavefunction.

Crucially, the longitudinal wavevector $k_z = \sqrt{k_0^2 n^2 - k_\perp^2}$ is dependent on both real space coordinates (x, y) and reciprocal space coordinates (k_x, k_y) , and includes any modification to the wavelength inside the medium via the refractive index $n(x, y)$.

Equation (8) can take advantage of the Fast Fourier Transform (FFT) for efficient computation

$$\psi(x, y, z + \Delta z) = \mathcal{F}^{-1} \left[e^{ik_z(x, y) \Delta z} \cdot \mathcal{F}\{\psi(x, y, z)\} \right] \quad (9)$$

and is implemented in a slice-by-slice manner akin to multislice. When the refractive index is constant across the transverse plane, WPM reduces to the Angular Spectrum of Plane waves method of propagation.

Although the WPM is accurate to high angles, its main drawback is that for each unique refractive index value in the slice, a unique propagator must be computed. Without any modification, the order of computational complexity is $O(N^{2D})$ where N is the number of grid points, and D is the number of transverse dimensions. Such a computational cost is prohibitive, as a an object slice that is discretised into for example 100 x 100 pixels, and has a unique refractive index value at each pixel, would require 10,000 unique Fourier transforms per slice for one propagation step.

[?] proposed a modification to the WPM that reduces the computational complexity from $O(N^{2D})$ to $O(MN^D \log N)$, where M is the number of "unique refractive" index values in the slice. By applying a mask $I_m(x, y)$ that selects pixels with the same refractive index n_m , the propagation step can be written as a sum over M unique propagations. However, in the case of electron microscopy multislice this algorithm poses two problems: First, applying a mask to the propagator negates the possibility to differentiate through the propagation step, which is a necessary requirement to use the WPM as a forward model to solve inverse problems via gradient descent, as is the case in ptychography. Second, in the case of electron microscopy, the refractive index is continuous function of the potential $V(\mathbf{r})$, and thus there are usually not very many "unique" refractive index values in a slice. [?] attempted to overcome both of these problems by introducing a continuous binning step in a binary refractive index medium.

At this point it is pertinent to highlight why one cannot simply use the Angular Spectrum Method (ASM) to reintroduce high angular scattering into the multislice formulation. In the process of deriving the multislice equation, one must linearise the square root operator in the propagation step by splitting the Laplacian and refractive index terms. This process is performed by Taylor expanding the square root operator, and neglecting higher order terms. This linearisation step is what introduces the paraxial approximation into the multislice method, and thus prevents the ASM multislice method from being accurate at high angles. The WPM however, does not require this linearisation step, and thus retains high angle accuracy. In order to reintroduce higher angle propagation in the split-step multislice propagation framework, one must retain higher order terms in the Taylor expansion of the operator, and in the optics literature these methods are known as wide-angle or Padé approximants. As far as the author is aware, these wide-angle multislice methods have not been applied to electron microscopy simulations.

4 Binning of refractive index values for efficient WPM multislice

Inspired by their implementation, we have adapted the WPM multislice algorithm to use a refractive index binning strategy which can handle the continuous nature of the refractive index in electron microscopy multislice potentials.

We discretize the refractive index range $[n_{\min}, n_{\max}]$ of the current slice into N_{bins} discrete reference values, denoted as $\{n_{\text{ref}}^{(k)}\}_{k=1}^{N_{\text{bins}}}$.

In a typical multislice potential, many of the refractive index values will be very close to 1 (vacuum region), while the refractive index values of an atom will concentrate in a narrow range slightly above 1. Since there are likely to be many more refractive index values near the value of the vacuum region, we employ a quadratic distribution for the reference indices rather than using linearly spaced bins in order to allocate more bins near $n = 1$. The reference indices are distributed according to a quadratic function determined by the parameter γ , with $\gamma = 2.0$ for quadratic spacing:

$$n_{\text{ref}}^{(k)} = n_{\min} + (n_{\max} - n_{\min}) \left(\frac{k}{N_{\text{bins}} - 1} \right)^\gamma$$

which allows for denser binning near the vacuum refractive index when $\gamma > 1$.

Next, for a propagation step Δz , we compute a bank of propagated fields $u_k(x, y)$. Each field u_k corresponds to the diffraction of the input field $u_{\text{in}}(x, y)$ assuming a homogeneous medium with refractive index $n_{\text{ref}}^{(k)}$. This is performed efficiently via batch FFT operations using the angular spectrum kernel:

$$u_k(x, y) = \mathcal{F}^{-1} \left(\mathcal{F}(u_{\text{in}}) e^{i\Delta z \sqrt{(k_0 n_{\text{ref}}^{(k)})^2 - k_x^2 - k_y^2}} \right)$$

To evaluate the field at a specific spatial coordinate (x, y) where the local refractive index is $n(x, y)$, we identify the two bounding reference bins such that $n_L \leq n(x, y) < n_R$, where $n_L, n_R \in \{n_{\text{ref}}^{(k)}\}$. We propose a differentiable interpolation scheme, by defining a local interpolation weight

$$w_{\text{raw}} = \frac{n(x, y) - n_L}{n_R - n_L}, \quad w_{\text{raw}} \in [0, 1]$$

To ensure smooth gradients and minimize discretization artifacts, we map this linear weight through a smoothstep activation function $S(w)$, similar to the spatial partitioning technique demonstrated by [?] via a cubic Hermite interpolation function:

$$w = S(w_{\text{raw}}) = 3w_{\text{raw}}^2 - 2w_{\text{raw}}^3$$

While linear interpolation would satisfy the requirement for differentiability, it could result in discontinuities in the gradient at the bin boundaries in a gradient based optimisation scheme.

The final field at the next step $u(x, y, z + \Delta z)$ is computed as the weighted superposition of the fields propagated at the bounding indices:

$$u(x, y) = (1 - w) \cdot u_L(x, y) + w \cdot u_R(x, y)$$

This formulation ensures that the propagation model is fully differentiable with respect to the refractive index map $n(x, y)$, enabling inverse reconstruction of potentials via automatic differentiation, and that only a limited number of FFTs (equal to N_{bins}) are required per propagation step, significantly reducing the computational cost compared to a naive WPM implementation. The accuracy of this binned WPM multislice method depends on the number of bins N_{bins} chosen and will change on a case by case basis depending on the refractive index distribution of the sample. One can mitigate concern about this potential source of error by performing a test simulation, and doubling the number of bins until no difference is observed in the output wavefunction.

In general, the WPM will need a GPU for efficient computation, and we have found that N_{bins} between 64 and 512 was sufficient for the simulations presented in this work. This is a marked improvement on the naive case, as it enables the WPM simulation to compute a forward simulation with many slices and a larger pixel size of 512×512 , or 1024×1024 pixels in a reasonable time frame (seconds to minutes).

5 Verification of WPM multislice accuracy at high angles

To verify the accuracy of our WPM multislice implementation, we have recreated the simulation presented in figure 3 of [?], which compares the amplitude of a central and high angle (135 mrad) diffracted beam after propagation through a thick gold sample. We can see that for the central beam, all methods (MS, and WPM) agree. For the high angle beam, at 135 mrad, the amplitude of the diffracted MS solutions deviates from the "ground truth" ODE solution as the thickness of the unit-cell increases. However, the WPM solution follows the trend of the ground truth ODE solution more closely, demonstrating that the WPM is more accurate at high angles than the traditional MS method, while maintaining the efficiency and larger step size of the FFT based multislice method (with two slices per unit-cell), provided one can use a GPU for computation of the many FFTs required per slice.

6 Thick Sample Comparison of WPM and MS in STEM simulation

In order to test the WPM multislice method in a more realistic scenario, we have performed a STEM simulation of a thick sample of gold (100 nm), tilted at 30 degrees relative to the incident electron beam. We compare the WPM multislice method to the traditional Fresnel multislice method. One can

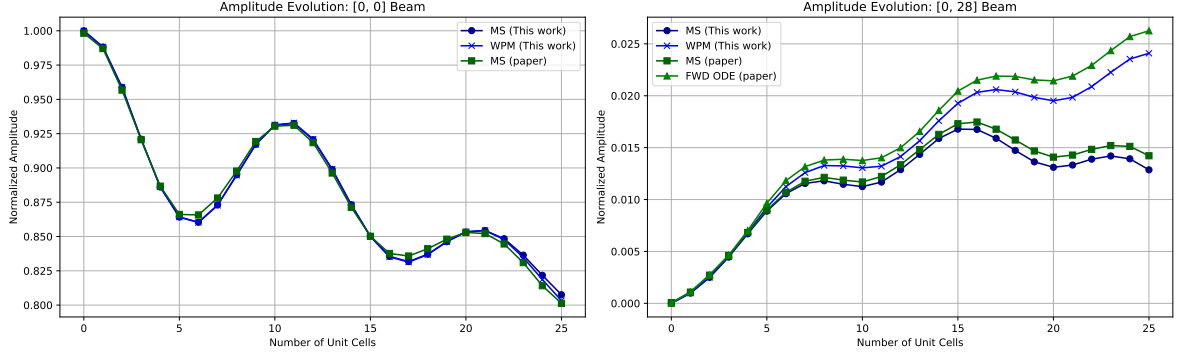


Figure 1: Recreation of figure 3 in [?] showing the amplitude of the central and high angle (135 mrad) diffracted beams, after propagation with our multislice and wave propagation methods. Plotted in green is the "ground truth" ODE solution, and the multi-slice solution of to the Klein-Gordon equation. In blue is the Angular Spectrum Multislice (ASMS) solution and the Wave Propagation Method (WPM) solution

